



WHAT IF SOMEONE HACKED THE UK POWER GRID RIGHT NOW?

AI-Based Cyberattack Detection in Smart Power Grid Systems

Real-time AI detection, classification, and explanation of cyberattacks in power systems

Hei Yu Chau

2370932



94.6%
ATTACK DETECTION RECALL



10,536 / 11,133
attacks caught

1. THE PROBLEM



Modern power grids are **increasingly vulnerable** to cyberattacks.



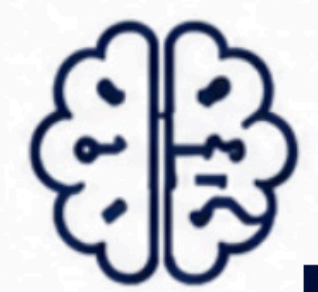
The 2015 Ukraine attack left **225,000 people** without electricity.



Most AI systems are **black-box models** with no human control.

2. PROPOSED SOLUTION

Detect, classify, explain, and decide — with human verification.



Detect
Attacks in Real Time



Classify
Attack Types Accurately



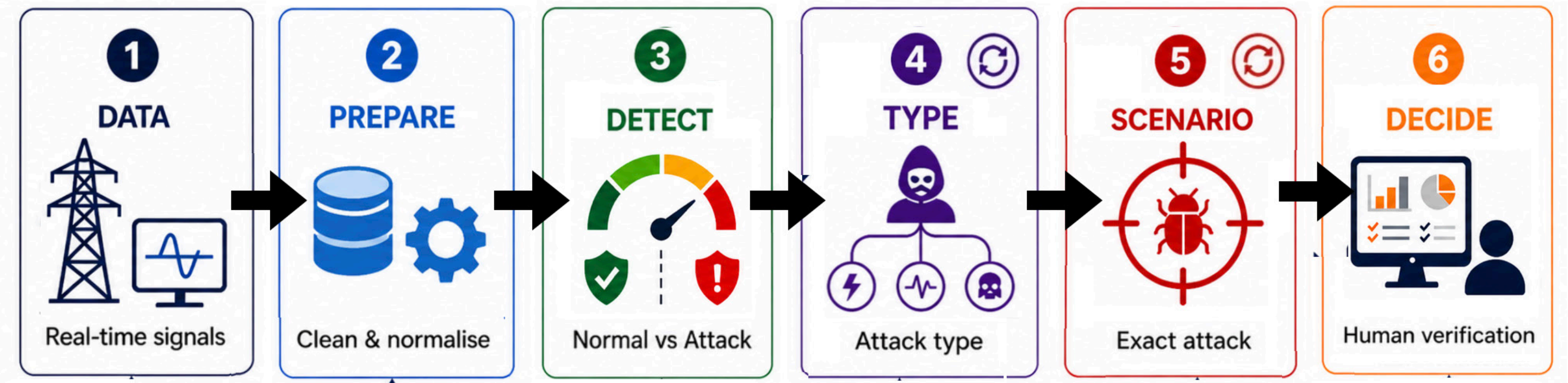
Explain
Decisions with Key Features



Decide
Human-in-the-Loop Verification

AI + Explainability + Human-in-the-Loop = Safer Power Grids

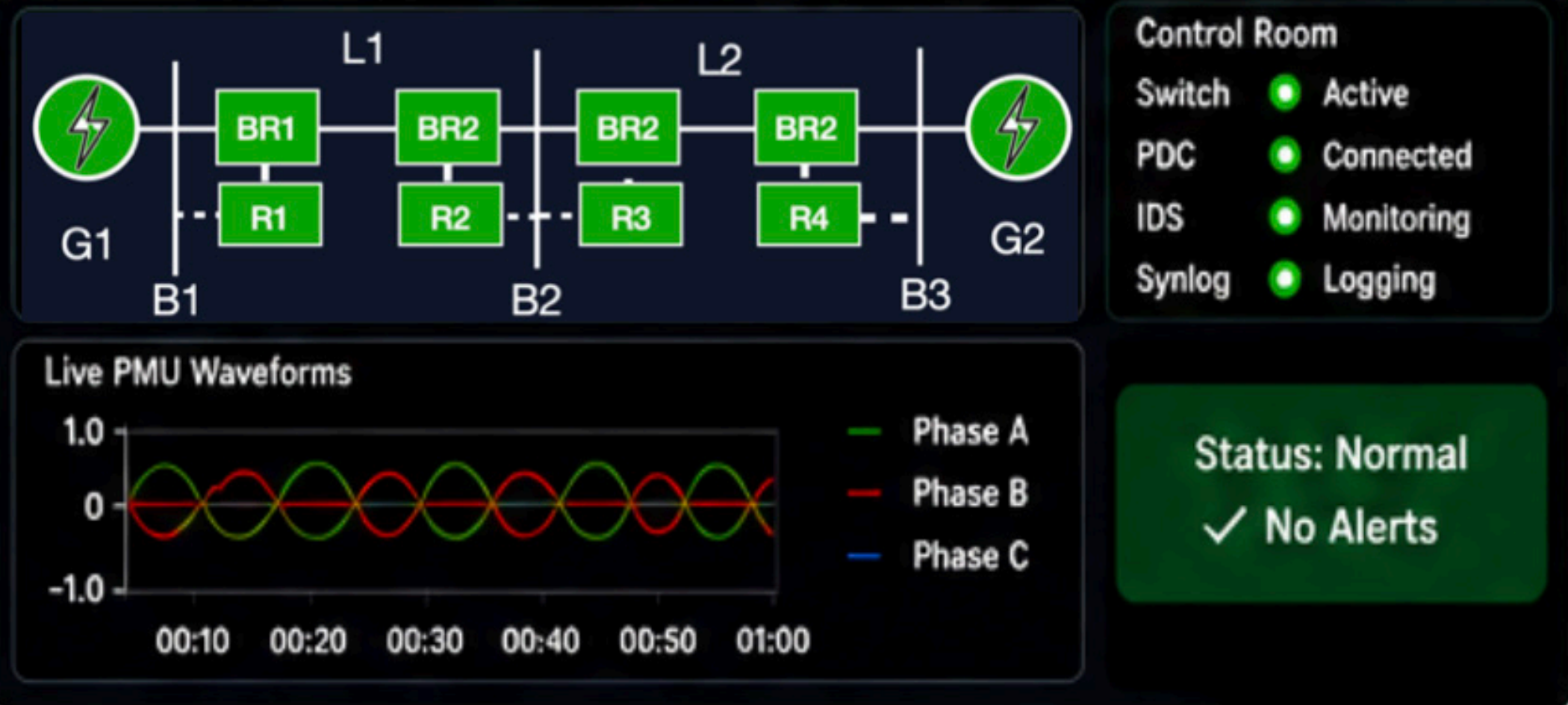
3. SYSTEM PIPELINE (End-to-End Workflow)



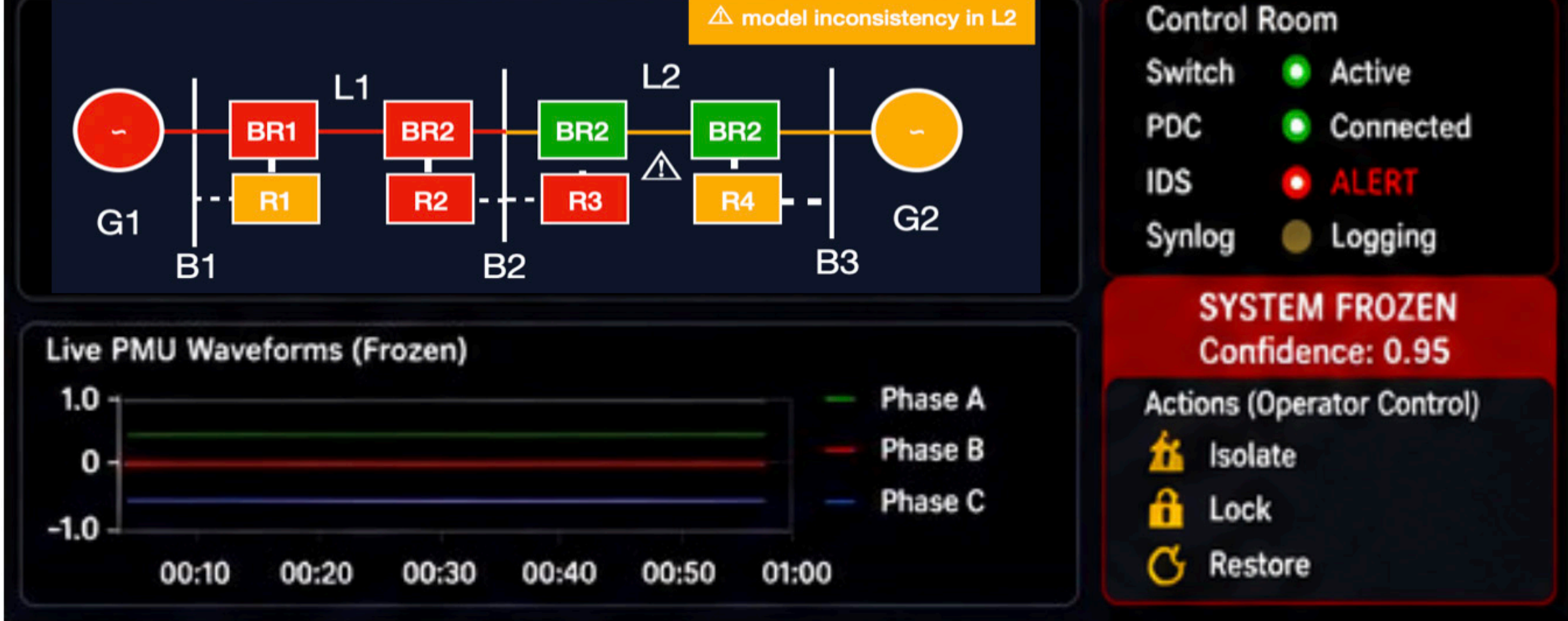
FALLBACK MECHANISM: Uncertain result → Re-check (safe route)

4. DASHBOARD OVERVIEW (Debug / Live)

NORMAL OPERATION – SYSTEM HEALTHY



ATTACK DETECTED – SYSTEM FROZEN



5. HUMAN-IN-THE-LOOP

When an attack is detected, the system **pauses**. The operator reviews **explainable insights** before action.



Human-verified decisions ensure safety in critical infrastructure.



Real-time alerts with **confidence scores**.



Explainable outputs (key features driving decisions).

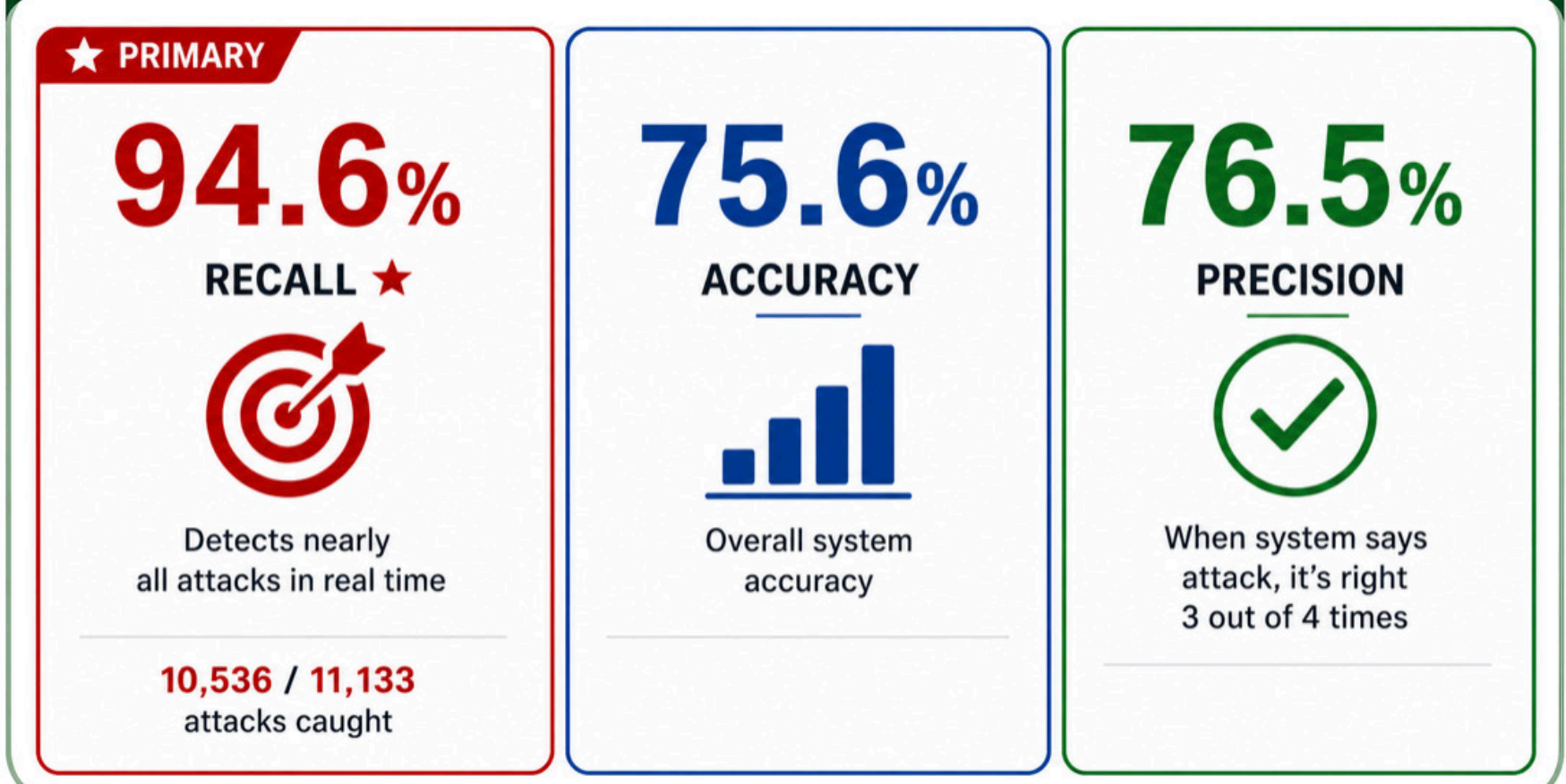


Decision support for response actions (isolate, restore, ignore).



Response time: **< 1 second** per event.

6. KEY PERFORMANCE (Test Set)



7. SYSTEM FEATURES

- Real-time detection + confidence score
- Live PMU monitoring (freeze on alert)
- Grid localisation (fault source)
- Explainable AI (key features + decision path)
- Human-in-the-loop decision control
- Debug & Live modes

8. TECHNICAL NOTES

- DATA:** 78k | PMU/SCADA | LR + RF
- TRADE-OFF:** High recall | false alarms
- SETUP:** Simulated
- LIMIT:** Human delay

9. KEY INSIGHT



The real challenge is not detection — but **distinguishing cyberattacks** from **real physical faults**.



SCAN ME

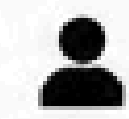
Demo + Code + Full System Details



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